PULLBACK OF DESCENT DATA ALONG AN FPQC REFINEMENT OF A FAMILY IS FULLY FAITHFUL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a commutative diagram of four categories of descent data; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms of families of maps with fixed target and descent data relative to a family. Sites, Definition sites-definition-morphism-coverings, invoked at proof lines 8959-8960, together with the same chapter's Lemma lemma-pullback-family invoked at proof lines 8971 and 9007-9008. Fixes the objects the functor acts on and whose commutativity with the coproduct description is used.
- A disjoint union of fpqc coverings is an fpqc covering. Topologies, Lemma topologies-lemma-disjoint-union-is-fpqc-covering, invoked at proof lines 8984-8986. Turns the hypothesis on each V_j into an fpqc covering of V_j by the corresponding sub-coproduct.
- Recognition criterion for fpqc coverings, part (3). Topologies, Lemma topologies-lemma-recognize-fpqc-covering part (3), invoked at proof lines 8993-8996. Converts the existence of finite affine covers of affine opens of Y into the statement that the single morphism g is an fpqc covering.
- Descent data for a family agree with descent data for the associated coproduct morphism, and pullback along an fpqc covering of schemes is fully faithful. Same chapter, Lemma lemma-family-is-one at proof lines 9003-9005 and Lemma lemma-fully-faithful at proof lines 9009-9011. Makes the two vertical arrows of the comparison square equivalences and supplies full faithfulness of the top horizontal arrow.

Source reference. Original lemma 023X: sources/stacks-descent/pinned-original-source.tex, source lines 8467–8467. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0241: sources/stacks-algebraic/pinned-original-source.tex, source lines 2746–2746. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02VS: sources/stacks-descent/pinned-original-source.tex, source lines 8570–8570. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 00VT: sources/stacks-chow/tags.txt, source lines 1159–1159. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 040I: sources/stacks-chow/tags.txt, source lines 5257–5257. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03L7: sources/stacks-chow/tags.txt, source lines 4703–4703. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{U} = \{U_i \to S\}_{i\in I}$, and $\mathcal{V} = \{V_j \to S\}_{j\in J}$, be families of morphisms with target S. Let $\alpha : I \to J$, $id : S \to S$ and $g_i : U_i \to V_{\alpha(i)}$ be a morphism of families of maps with fixed target, see Sites, Definition 00VT. Assume that for each $j \in J$ the family $\{g_i : U_i \to V_j\}_{\alpha(i)=j}$ is an fpqc covering of V_j. Then the pullback functor*

$$\textit{descent data relative to } \mathcal{V} \longrightarrow \textit{descent data relative to } \mathcal{U}$$

of Lemma 02VS is fully faithful.

Proof. Consider the morphism of schemes

$$g : X = \coprod\nolimits_{i\in I} U_i \longrightarrow Y = \coprod\nolimits_{j\in J} V_j$$

over S which on the ith component maps into the $\alpha(i)$th component via the morphism $g_{\alpha(i)}$. We claim that $\{g : X \to Y\}$ is an fpqc covering of schemes. Namely, by Topologies, Lemma 040I for each j the morphism $\{\coprod_{\alpha(i)=j} U_i \to V_j\}$ is an fpqc covering. Thus for every affine open $V \subset V_j$ (which we may think of as an affine open of Y) we can find finitely many affine opens $W_1, \ldots, W_n \subset \coprod_{\alpha(i)=j} U_i$ (which we may think of as affine opens of X) such that $V = \bigcup_{i=1,\ldots,n} g(W_i)$. This provides enough affine opens of Y which can be covered by finitely many affine opens of X so that Topologies, Lemma 03L7 part (3) applies, and the claim follows. Let us write $DD(X/S)$, resp. $DD(\mathcal{U})$ for the category of descent data with respect to X/S, resp. $\mathcal{U}$, and similarly for Y/S and $\mathcal{V}$. Consider the diagram

$$\begin{array}{ccc} DD(Y/S) & \longrightarrow & DD(X/S) \\ \uparrow{\scriptstyle \text{Lemma } 023X} & & \uparrow{\scriptstyle \text{Lemma } 023X} \\ DD(\mathcal{V}) & \longrightarrow & DD(\mathcal{U}) \end{array}$$

This diagram is commutative, see the proof of Lemma 02VS. The vertical arrows are equivalences. Hence the lemma follows from Lemma 0241 which shows the top horizontal arrow of the diagram is fully faithful. □